

RULES AND REGULATIONS FOR THE WRITTEN EXAMS
AT THE FACULTY OF ENGINEERING AND NATURAL SCIENCES

Please read this document carefully, sign the bottom part, and include this form with your solution sheet.

- The exam duration (start-end times) will be written on your exam sheet or will be announced, you should adhere to these times.
- You cannot leave the examination room in the **first 30 minutes** of the exam.
- You are **NOT allowed** to take the exam once a student has left one of the exam rooms.
- Talking with another student during the exam for any reason (asking for an explanation, for a calculator or eraser or any other object) is **NOT allowed**.
- If you **GIVE** to or **RECEIVE** from a student any item (eraser, calculator, pen, etc.) during the exam, this will be taken as a sign of cheating.
- **TURN OFF** your CELLPHONES AND SMART DEVICES; and **PLACE THEM away from you** such that those **CANNOT BE SEEN** from outside.
- **HAVING a CELLPHONE or a SMART DEVICE** that is **VISIBLE FROM OUTSIDE** during the exam is **NOT ALLOWED**. Interacting with a cellphone in any way (EVEN IF IT IS TURNED OFF) will be taken as a sign of cheating.

In case of a sign of cheating, or if you do not obey the above rules, there will be a disciplinary action taken towards you based on the STUDENT DISCIPLINARY REGULATIONS of YÖK Items 2.5, 2.7, and 2.8.

I understand the above-mentioned rules and regulations, and I hereby accept them, and any consequences following my actions during the exam.

Student Full Name:

Student Number:

Course Code and Title:

AIN-2002 INTRODUCTION TO DATA SCIENCE

Signature:

Date:

Q.1	Q.2	Q.3	Q.4	TOTAL
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Q.1) (20 pts) You are a data analyst at a national retail chain and are evaluating the effectiveness of a new targeted advertising campaign. To assess its impact, you compare sales data from two different stores:

- Store A (did not receive the new campaign)
- Store B (received the campaign for 30 days)

You collected the following information:

- Store A (Control Group):
Sample size: 40 days
Sample mean daily sales: \$495
Sample standard deviation: \$90
- Store B (Treatment Group):
Sample size: 30 days
Sample mean daily sales: \$550
Sample standard deviation: \$100

A) Define the appropriate null and alternative hypotheses for this analysis. (5pts)

B) Choose and justify the correct statistical test. Then, perform the test at a 0.05 significance level. (5pts)

C) Calculate the degrees of freedom using the Welch–Satterthwaite equation. (5pts)

D) Based on your test result, state whether there is statistical evidence that the advertising campaign had an effect on daily sales. Explain your conclusion. (5pts)

Q.2) (25 pts) You are evaluating a multiple linear regression model designed to predict stress levels among young adults (aged 18–25) based on social media usage (hours/day) and sleep duration (hours/night). The dataset provided includes:

- Actual (measured) stress levels
- Predicted stress levels from the model
- Residuals (errors): actual – predicted
- Usage patterns (e.g., high social media hours and low sleep)

Using the data table provided in the exhibit:

a) (5pts) Compute the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). Explain what each metric represents.

b) (5pts) Calculate the coefficient of determination (R^2) and explain its interpretation in the context of this study?

c) (5pts) Based on the residuals in the dataset, identify whether the model suffers from heteroscedasticity. Justify your answer using variance patterns?

d) (5pts) The model seems to underpredict stress levels for individuals who use social media for *more than 6 hours/day*. Describe how you could detect and visualize this issue, and what actions could be taken to improve the model.

e) (5pts) If the RMSE is relatively low but R^2 is also low, what could be a possible explanation? Explain this paradox with examples from the data.

Q.3) (25pts) Please refer to the code exhibit and answer the related questions:

```
library(ggplot2)
library(dplyr)
library(FactoMineR)
library(factoextra)

data <- read.csv("https://archive.ics.uci.edu/ml/machine-learning-databases/wine-
quality/winequality-red.csv", sep = ";")

data$quality <- as.factor(data$quality)
head(data)

run_anova <- function(var) {
  formula <- as.formula(paste(var, "~ quality"))
  aov_result <- aov(formula, data = data)
  summary(aov_result)
}

numeric_vars <- names(data)[sapply(data, is.numeric)]
numeric_vars <- setdiff(numeric_vars, "quality")
anova_results <- lapply(numeric_vars, run_anova)
names(anova_results) <- numeric_vars

wine_pca <- PCA(scale(data[, numeric_vars]), graph = FALSE)

fviz_eig(wine_pca, addlabels = TRUE)

fviz_pca_ind(wine_pca,
             geom.ind = "point",
             col.ind = data$quality,
             palette = "jco",
             addEllipses = TRUE,
             legend.title = "Wine Quality")
```

a) (5pts) What is the purpose of converting quality into a factor variable in the context of ANOVA and PCA visualizations?

b) (5pts) Interpret the purpose of the `run_anova` function. What would a low **p-value** for "`alcohol ~ quality`" imply?

c) (5pts) Why is the `scale()` function applied before running PCA? What might happen if it were omitted?

d) (5pts) Examine the PCA scree plot. If the first two components explain less than 70% of the variance, what might be a good strategy for improving the dimensionality reduction?

e) (5pts) Based on both the ANOVA results and PCA biplot, evaluate whether alcohol and volatile acidity are good predictors for wine quality. Justify your answer based on variance and grouping?

Q.4) (25pts) Please supply the convenient R code for the following requirements

a) (5pts) Write an R function named **poly_eval** that takes a numeric vector x and returns the evaluation of the function $f(x)=x^3-2x^2+x-5$. Then, use it to compute the values for $x = -5$ to 5 .

b) (5pts) Plot the function from part (a) using ggplot2, showing points and a smooth line. Label the axes and add a title?

c) (5pts) Simulate **1000 biased coin tosses**, where "Heads" has probability **0.3** and "Tails" has probability **0.7**. Use a table to summarize the results?

d) (5pts) Generate a numeric vector: $x = c(3.5, 4.2, NA, 6.1, 7.3, NA, 5.8)$. Find the mean and median, handling the missing (NA) values correctly?

e) (5pts) Generate 200 values from a normal distribution with mean=100 and sd=15. Then, create a horizontal boxplot with colored outliers.